

- Q.29 Describe the function, construction, and working principle of telescopic shock absorbers. How do different types of shock absorbers, such as twin-tube and mon-tube designs, influence ride quality and handling characteristics?
- Q.30 Explain the concept of specific gravity in relation to battery electrolyte. How does temperature affect the specific gravity of electrolyte?
- Q.31 Discuss the constructional details of a lead-acid cell battery, highlighting the role and arrangement of components such as electrodes, electrolyte, separators, and casing. How do these components contribute to the battery's overall function and performance?
- Q.32 Explain the function and operational details of a dynamo, including its role in generating electrical power for automotive systems.
- Q.33 Discuss the construction and working principles of an alternator, emphasizing its role in generating electrical power for charging the battery in an automotive system.
- Q.34 Explain the typical wiring diagram of an automobile incorporating an alternator.
- Q.35 Outline the process of charging a battery using an alternator. Detail the key components involved in the charging process.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Explain the various factors affecting tyre life.
- Q.37 Explain the working of a sliding mesh gear box with 3 forward and 1 reverse gear.
- Q.38 Explain the construction & working of Differential.

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6th Sem / Mech, GE, Mech. Engg. (Fabrication Tech.)

Subject:- Automobile Engineering

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Which of the following describes the layout of a chassis?
a) Types of engines
b) Types of drives
c) Classification of automobiles
d) Structure of the frame
- Q.2 What technology is associated with 4-cylinder engines featuring twin cam 16 valves?
a) Single overhead cam b) Double overhead cam
c) Single camshaft d) No camshaft
- Q.3 Which transmission system is known for its Torque converter and overdrive features?
a) Sliding mesh gearbox
b) Constant mesh gearbox
c) Automated Manual Transmission
d) Continuously Variable Transmission
- Q.4 What is the function of a universal joint in a vehicle's propeller shaft?
a) Transfers power from the engine to the wheels
b) Transmit torque at different angles
c) Balances the vehicle's weight
d) Controls the gear shifting process
- Q.5 Which type of steering gear is commonly found in power steering systems?
a) Worm and wheel
b) Rack and pinion
c) Hydraulic clutch
d) Cone clutch

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- Q.6 What is the purpose of the Ackerman and Davis Steering Mechanism?
- Steering wheel alignment
 - Power steering adjustment
 - Improving steering efficiency
 - Braking system calibration
- Q.7 Which braking system is known for its Anti-lock Brake System (ABS)?
- Mechanical brake
 - Hydraulic brake
 - Air brake
 - Vacuum brake
- Q.8 What component regulates the voltage and current in a dynamo?
- Regulators
 - Cutout
 - Alternator
 - Integrated starter-alternator
- Q.9 Which type of battery is commonly used in electric and hybrid vehicles?
- Lead-acid cell
 - Lithium-ion
 - Nickel-metal hydride
 - Alkaline
- Q.10 What is the effect of temperature of the specific gravity of electrolyte in a lead-acid battery?
- Increases
 - Decreases
 - Remains constant
 - Fluctuates randomly

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 What is the function of a clutch in a vehicle.
- Q.12 The primary purpose of a master cylinder in a braking system is to control brake fluid _____.
- Q.13 Wheel balancing and alignment enhance vehicle _____.
- Q.14 The introduction of electric and hybrid vehicles is not significant in the automobile industry. (True/False)
- Q.15 Which type of battery is commonly used in electric and hybrid vehicles?
- Q.16 What is primary function of a shock absorber in a suspension system?
- Q.17 Which braking system is known for its Anti-lock Brake System (ABS)?

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- Q.18 What is the purpose of the Ackerman and Davis Steering Mechanism?
- Q.19 What is the effect of temperature on the specific gravity of electrolyte in a lead-acid battery?
- Q.20 Which type of gearbox is known for its synchronized gear shifting?

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Describe the classification of automobiles based on their body type, size, and purpose, providing examples for each classification.
- Q.22 Differentiate between front-wheel drive, rear-wheel drive, and four-wheel drive systems in vehicles, discussing their advantages and applications.
- Q.23 Explain the functions of a clutch in an automobile and discuss the constructional details of single plate clutch with neat diagram.
- Q.24 Describe the different types of rear axles and rear axle drives used in automobiles, highlighting their respective advantages and applications.
- Q.25 Discuss the importance of toe-in, toe-out, camber, caster, and kingpin inclination in wheel balancing and alignment and their effects on vehicle handling and tyre wear.
- Q.26 Discuss the Ackerman and Davis Steering Mechanism, describing how it ensures proper wheel alignment during turns and its significance in improving vehicle stability and tyre wear.
- Q.27 Explain the components and functions of a master cylinder and wheel cylinder in a braking system, discussing their roles in transmitting hydraulic pressure to engage the brakes.
- Q.28 Describe the constructional details and working principles of hydraulic brake system, highlighting their relative merits and demerits in terms of braking efficiency and maintenance requirements.

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